12 · Case 8, part 2: rheumatoid arthritis, cortisol, and Shannon

The words of this track, as it says them. Bold brown is the examiner.

Exam case 8, continued. Rheumatoid arthritis, cortisol, and what to tell Shannon.

Four marks. What occurs in rheumatoid arthritis?

The hook: autoimmune, fast, both sides, whole body.

It's autoimmune: the immune system attacks the joints. The onset is relatively rapid, over weeks to months, and it can begin at any time in life. The joints are painful, swollen and stiff, often small and large joints on both sides of the body: both hands, both wrists, the elbows, the balls of the feet. Morning stiffness lasts longer than an hour. And there are whole-body symptoms: frequent fatigue, and a general feeling of being ill.

One mark. What is the other type of arthritis called?

Osteoarthritis. The wear and tear one.

Four marks. What is the difference between osteoarthritis and rheumatoid arthritis?

Her table has four rows: the cause, the onset, the joints, and the whole body.

The cause. Rheumatoid is autoimmune. Osteoarthritis is wear and tear.

The onset. Rheumatoid can begin at any time in life, and comes on over weeks to months. Osteoarthritis usually begins later in life, slowly, over years.

The joints. In rheumatoid, they're painful, swollen and stiff, on both sides, small and large, with morning stiffness of more than an hour. In osteoarthritis there's aching and tenderness with little or no swelling, usually in one set of joints, like the fingers, thumbs, hips, knees or spine, and morning stiffness of less than an hour.

The whole body. Rheumatoid has whole-body symptoms: fatigue, and feeling ill. Osteoarthritis doesn't.

Two marks. Where is cortisol produced and secreted from in the body?

In the adrenal glands: paired, pyramid-shaped glands on top of the kidneys. Specifically, the adrenal cortex, the outer part. Cortisol is its main glucocorticoid, and its release is stimulated by ACTH, from the anterior pituitary.

Four marks. What are the functions of cortisol, and how is this related to the stress response?

The hook: more sugar, steady blood volume, and less inflammation.

Cortisol increases blood glucose, by forming glucose from non-carbohydrate sources, which is gluconeogenesis. It also raises the blood's fatty acids and amino acids. It prepares the

body to resist chronic, long-term stress, by keeping blood sugar relatively constant, and maintaining blood volume. Maintaining blood volume keeps the tissues perfused, so oxygen and nutrients are there to deal with the stress. And it reduces inflammation, so less swelling, redness and pain, and it suppresses the immune system.

Two marks. How is cortisol going to help Shannon's rheumatoid arthritis?

Two ways. It reduces inflammation: less swelling, redness and pain in her joints. And it suppresses the immune system, which matters, because rheumatoid arthritis is autoimmune: an overactive immune system attacking her joints. Prednisone is the synthetic version, doing the same job.

Two marks. What might you say to Shannon about her concerns?

Her own words from the session: rheumatoid arthritis is an autoimmune condition, and it can hit at any age. So Shannon is not too young for it. It's not the wear and tear of osteoarthritis, which usually begins later in life.

Once more. Rheumatoid arthritis, and cortisol.

Rheumatoid arthritis is autoimmune. It comes on over weeks to months, at any age, affects joints on both sides with morning stiffness over an hour, and brings whole-body fatigue. Osteoarthritis is wear and tear: later in life, over years, one set of joints, little swelling, stiffness under an hour, and no whole-body symptoms. Cortisol comes from the adrenal cortex, released by ACTH. It raises blood glucose by gluconeogenesis, keeps blood volume and perfusion up through long-term stress, reduces inflammation, and suppresses the immune system, which is how it helps Shannon. And tell her: it's autoimmune, and it can hit at any age.